0.1 Exact residual certificates for supplied packets

Source item 9.2; module DiscoveryCertificate

Let (P_a) be a complete orthogonal projector system on a finite complex coordinate space. The projectors need not be coordinate masks. For an actual matrix A, define

$$\Pi(A) = \sum_a P_a A P_a, \qquad E = A - \Pi(A).$$

Reconstruction gives the exact off-diagonal residual

$$E = \sum_{b \neq a} P_b A P_a.$$

Hilbert–Schmidt orthogonality. For distinct ordered packet pairs $(b, a) \neq (d, c)$, adjoint multiplication and projector orthogonality give

$$\operatorname{tr}\big((P_b A P_a)^* P_d B P_c\big) = 0.$$

If $b \neq d$, the central projector product vanishes. Otherwise $a \neq c$, and cyclicity of trace brings $P_c P_a$ together. This is proved for arbitrary input matrices A, B.

Lean identifies the actual Frobenius norm with its trace expression,

$$\|M\|_{\mathrm{F}}^2 = \operatorname{Re}\operatorname{tr}(M^* M),$$

then expands the finite sum and removes each cross term. The result is

$$\boxed{\|A - \Pi(A)\|_{\mathrm{F}}^2 = \sum_{b \neq a} \|P_b A P_a\|_{\mathrm{F}}^2 .}$$

No Pythagorean equality for the target residual is assumed.

Operator norm. In a separate scoped matrix norm, the triangle inequality gives

$$\boxed{\|A - \Pi(A)\|_2 \leq \sum_{b \neq a} \|P_b A P_a\|_2 .}$$

Here $\|\cdot\|_2$ is the Euclidean induced operator norm, not the Frobenius norm. The formal declarations use distinct norm instances, preventing the two certificates from being conflated. Self-adjointness is used for Frobenius orthogonality; the sum estimate itself requires only the complete packet decomposition.

Discovery boundary. This result certifies the residual of a *supplied* decomposition. It does not construct useful projectors from an arbitrary matrix, prove that a searched decomposition is optimal, or guarantee that its application is profitable on a GPU. The simultaneous spectral-decomposition input remains separately recorded as classical.